Link to the source code: GitHub Repository

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Problem Description

The problem considered in this project involves a set of nodes, each represented by three integer values: two coordinates (x,y)(x, y)(x,y) that define the node's position in a two-dimensional plane, and a cost value associated with the node. The objective is to select exactly 50% of all available nodes (if the total number of nodes is odd, the number of selected nodes is rounded up) and construct a Hamiltonian cycle — that is, a closed path visiting each selected node exactly once and returning to the starting node. The goal is to minimize the sum of two components: 1. The total length of the constructed cycle, and 2. The total cost of all selected nodes The distances between nodes are calculated using the Euclidean metric, rounded to the nearest integer.

Implemented Algorithms

```
Operator1(parent1, parent2)
    n = length(parent1)
    targetSize = n / 2

    edges1 = empty set
    edges2 = empty set

    for i = 0 TO n-1 do
        a = parent1[i]
        b = parent1[(i+1) mod n]
        add (a, b) to edges1

    for i = 0 TO n-1 do
        a = parent2[i]
        b = parent2[(i+1) mod n]
        add (a, b) to edges2

    visited = empty set
    subpaths = empty list of lists

    for i = 0 TO n-1 do
        start = parent1[i]

        if start in visited then
            CONTINUE

        path = [ start ]
        mark start as visited
        current = start

        while TRUE do
            nextIndex = index of current in parent1 + 1

            if nextIndex ≥ n then
                break

            next = parent1[nextIndex]
            edge = (current, next)

            if edge in edges1 AND edge in edges2 AND next not in
visited then
                append next to path
                mark next as visited
                current = next
            else
                break

        append path to subpaths

    allNodes = set of all nodes in parent1
    remove all visited nodes from allNodes

    currentSize = total number of nodes in subpaths

    while currentSize < targetSize AND allNodes is not empty do
        node = randomly selected element from allNodes
        append [node] to subpaths
        remove node from allNodes
        currentSize = currentSize + 1


    randomly shuffle subpaths

    offspring = empty list

    for each path in subpaths do
        if random number < 0.5 then
            reverse path
        append path to offspring

    RETURN offspring

GreedyRegretStart(instance, start_node):
    Select start_node
    Select second node with minimum (distance + cost)

    while selected nodes < K:
        for each unselected node v:
            compute best insertion cost
            compute second-best insertion cost
            regret = second_best - best
            score = regret - best_total_cost

        choose node with maximum score
        insert node at its best position

    return tour, inSelected[]

CleanPopulation(instance, population, targetSize):
    RemoveDuplicates
    if population too large:
        RemoveWorst
    if population too small:
        generate new LS-improved random solutions
    return population

GenerateInitialPopulation(instance, size):
    while population size < size:
        with probability 0.8:
            greedy regret construction
        else:
            random + local search
    remove duplicates
    return population

Main Loop

RunMethod(instance):
    population = GenerateInitialPopulation(instance, populationSize)

    start timer
    while time limit not exceeded:
        select random pairs of parents
        for each pair:
            offspring = crossover operator
            apply local search to offspring
            add offspring to population

        CleanPopulation(instance, population, populationSize)

    return best solution found
```

Results

Comparison of previous methods results

Instance A

Method	Best	Worst	Average
random	235308	292123	264677.17
nn_end	89198	123123	104012.785
nn_anywhere	71488	74410	72635.72
greedy cycle	72639	72639	72639.0
2-Regret Insertion	105852	123428	115474.93
Weighted Sum (α=1.00, β=1.00)	71108	73438	72130.85
greedy_intra:edges_start:greedy	70663	72648	71594.055
greedy_intra:edges_start:random	71564	76328	73528.81
greedy_intra:nodes_start:greedy	70687	73027	71739.395
greedy_intra:nodes_start:random	79677	99035	86909.13
steepest_intra:edges_start:greedy	70510	72614	71463.08
steepest_intra:edges_start:random	71246	78153	73699.995
steepest_intra:nodes_start:greedy	70626	73004	71615.93
steepest_intra:nodes_start:random	81322	95342	87939.335
steepest_intra:edges_start:random	71246	78153	73699.995
steepest_intra:edges_start:random with Cand mech.	73076	85487	77866.91
steepest_lm_intra:edges_start:random	71265	78247	73856.275
ILS_intra:edges_start:random	70901	71862	71303.60
steepest_multi_start_intra:edges_start:random	71149	72794	71299.05
LNS_no_search_intra:edges_start:random	69732	71868	70604.2
LNS_with_search_intra:edges_start:random	69397	71604	70453.9
Operator1	69875	70843	70518.45
Operator2	70154	72087	71630.65
Operator2_without_LS	70233	71965	71397.2

Hybrid evolutionary algorithm

Method	best	worst	average
GeneticWithRegret	69620	70145	69853.05

Instance B

Method	Best	Worst	Average
random	193234	234798	214279.235
nn_end	62606	77453	69764.43
nn_anywhere	49001	57324	51400.905
greedy cycle	50243	50243	50243.0
2-Regret Insertion	66505	77072	72454.77
Weighted Sum (α=1.00, β=1.00)	47144	55700	50918.82
greedy_intra:edges_start:greedy	46178	55471	50172.67
greedy_intra:edges_start:random	45537	51687	48154.865
greedy_intra:nodes_start:greedy	46373	55385	50347.525
greedy_intra:nodes_start:random	53417	71474	61453.295
steepest_intra:edges_start:greedy	45867	54814	49907.205
steepest_intra:edges_start:random	45903	51416	48195.845
steepest_intra:nodes_start:greedy	46371	55385	50201.47
steepest_intra:nodes_start:random	55686	71546	62955.885
steepest_intra:edges_start:random	45903	51416	48195.845
steepest_intra:edges_start:random with Cand. Mech.	45798	52670	48474.64
steepest_lm_intra:edges_start:random	45941	51587	48214.715
ILS_intra:edges_start:random	45320	46753	45463.3
steepest_multi_start_intra:edges_start:random	45405	46222	45767.4
LNS_no_search_intra:edges_start:random	44415	46341	45336.95
LNS_with_search_intra:edges_start:random	44012	45376	44511.30
Operator1	43924	45040	44757.4
Operator2	44244	45547	44894.6
Operator2_without_LS	44268	45724	44679.9

Hybrid evolutionary algorithm

Method	best	worst	average
GeneticWithRegret	44128	45207	44627.6

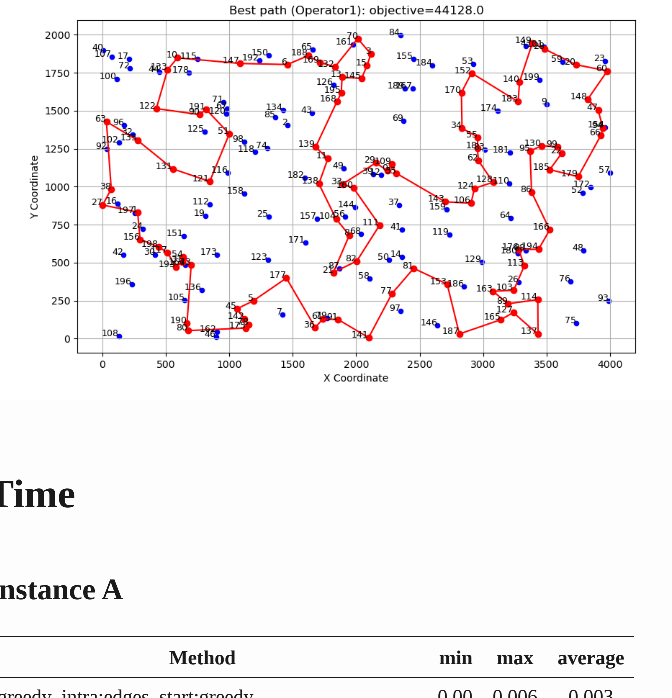
Paths visualization

Instance A

[](./best_path_Operator1_A.png)

Instance B

Operator 1



Time

Instance A

Method	min	max	average
greedy_intra:edges_start:greedy	0.00	0.006	0.003
greedy_intra:edges_start:random	0.062	0.099	0.080
greedy_intra:nodes_start:greedy	0.000	0.005	0.002
greedy_intra:nodes_start:random	0.066	0.118	0.085
steepest_intra:edges_start:greedy	0.001	0.005	0.002
steepest_intra:nodes_start:greedy	0.000	0.005	0.002
steepest_intra:nodes_start:random	0.049	0.142	0.071
steepest_intra:edges_start:random	0.039	0.050	0.045
steepest_intra:edges_start:random with Can. Mech.	0.008	0.019	0.011
steepest_lm_intra:edges_start:random	0.025	0.049	0.030
ILS_intra:edges_start:random	4.805	4.815	4.811
steepest_multi_start_intra:edges_start:random	4.443	5.99	4.805
LNS_no_search_intra:edges_start:random	4.810	4.891	4.840
LNS_with_search_intra:edges_start:random	4.812	4.881	4.832

New Methods Results

Method	min	max	average
GeneticWithRegret	-	-	as MSLS

Instance B

Method	min	max	average
greedy_intra:edges_start:greedy	0.0	0.01	0.004
greedy_intra:edges_start:random	0.064	0.102	0.081
greedy_intra:nodes_start:greedy	0.0	0.007	0.003
greedy_intra:nodes_start:random	0.058	0.107	0.082
steepest_intra:edges_start:greedy	0.001	0.01	0.004
steepest_intra:nodes_start:greedy	0.001	0.007	0.004
steepest_intra:nodes_start:random	0.048	0.086	0.063
steepest_intra:edges_start:random	0.038	0.054	0.045
steepest_intra:edges_start:random with CAn. Mech.	0.009	0.05	0.012
steepest_lm_intra:edges_start:random	0.020	0.053	0.035
ILS_intra:edges_start:random	4.499	4.515	4.503
steepest_multi_start_intra:edges_start:random	4.377	5.077	4.498
LNS_no_search_intra:edges_start:random	4.499	4.609	4.554
LNS_with_search_intra:edges_start:random	4.501	4.607	4.540

Method	min	max	average
GeneticWithRegret	-	-	as MSLS

Conclusions

In this assignment we explored influence of greedy regret initialization for the evolutionary algorithm, as it performed the best among all approaches. The results indicate that the hybrid evolutionary algorithm with greedy regret initialization outperforms other methods in terms of solution quality for both instances A and B. The best solutions obtained by this method are lower than those achieved by the other approaches, demonstrating its effectiveness in solving the problem at hand.